Measurement error & reproducibility

A quick visual reference for judging how trustworthy a measurement is

Measurement error

The difference between an observed value and the true value, made of random and systematic parts.

Reproducibility

How consistent results are across different raters, instruments, labs, or occasions.

Two error types, two consistency concepts

Random error

Scatters results both ways, averages out

Systematic error (bias)

Pushes results the same direction every time

Repeatability

Same method, same conditions, same operator

Reproducibility

Different conditions: rater, device, lab, time

Standard error of measurement, from the intraclass correlation (ICC)

$$SEM = SD \times \sqrt{(1 - ICC)}$$

ICC close to 1 means scores are mostly true differences between subjects, not noise

Accuracy versus precision

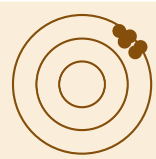
High precision (low random error)

Low precision (high random error)

High accuracy

Ideal

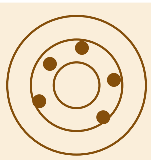
Tight cluster on the bull's-eye



Biased but consistent

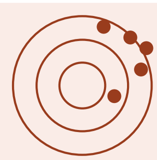
Tight cluster, off-target

Low accuracy



Correct on average

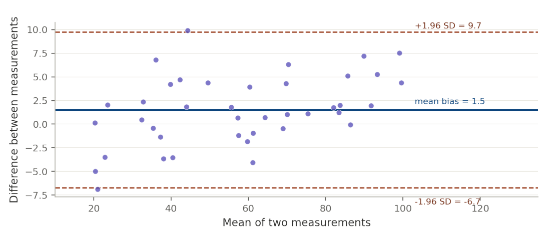
Scattered but centred



Worst case

Scattered and off-target

Bland-Altman plot: checking agreement between two measurement methods



Quick reference: a flat band of points hugging the bias line means good agreement; a fanning pattern means error grows with magnitude.